

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration : 30 Minutes

QUESTIONS: 20

Total Marks:20

Annexure A

MCQ Test

Code of Conduct:

*I (J O Joanna Divine / 192572003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Joanna

Annexure A - MCQ Test Solutions

Q1. Answer: C Greedy Best-First Search selects nodes using only the heuristic value $h(n)$.

Q2. Answer: C A* uses $f(n)=g(n)+h(n)$.

Q3. Answer: B An admissible heuristic never overestimates and guarantees optimality. Q4. Answer: B Hill Climbing is a local search algorithm.

Q5. Answer: B Minimax is used in adversarial games.

Q6. Answer: B Using only estimated distance corresponds to Greedy Best-First Search. Q7. Answer: B GPS combines travelled cost and estimated remaining cost like A*.

Q8. Answer: B Exam scheduling is a Constraint Satisfaction Problem.

Q9. Answer: C Chess programs commonly use Minimax.

Q10. Answer: B An Online Search Agent learns while exploring.

Q11. Answer: B Admissibility ensures A* finds the optimal solution.

Q12. Answer: B Alpha-Beta pruning reduces evaluated nodes.

Q13. Answer: C Hill Climbing may get stuck in local maxima.

Q14. Answer: C A CSP solution must satisfy all constraints.

Q15. Answer: B Evaluation functions estimate state utility.

Q16. Answer: A $f(n)=7+5=12$.

Q17. Answer: B Overestimating heuristics make A* non-optimal.

Q18. Answer: C Nodes= $1+2+4+8+16=31$.

Q19. Answer: B Leaf nodes= $3^3=27$.

Q20. Answer: B Best-case Alpha-Beta complexity is $O(b^{d/2})$

RUBRICS FOR CALCULATIONS:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions